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ESI-LAG Made Easier with EZ-LAG

An HPE Juniper Networking Techpost

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Contents

1 ESI-LAG Made Easier with EZ-LAG 4
1.1 Introduction . . . 4
1.2 Topology . . . 4
1.3 How Does EZ-LAG Work? . . . 5
1.4 Basic EZ-LAG Configuration . . . 18
1.4.1 Peer 1 Configuration . . . 18
1.4.2 Peer 2 Configuration . . . 19
1.4.3 Host-1 Configuration Files (Centos 7) . . . 19
1.4.4 Host-2 Configuration (Ubuntu 20.04) . . . 20
1.5 Verifications . . . 21
1.6 Conclusion . . . 29
1.7 Useful links . . . 29
1.8 Glossary . . . 30
1.9 Acknowledgments . . . 30

List of Figures

1 Reference Fabric Topology 4
2 Reference Topology for EZ-LAG Configuration 5

List of Tables

1 Parameters in Services EVPN Configuration Hierarchy 8
2 Default and Auto-Derived Parameters 14

1 ESI-LAG Made Easier with EZ-LAG

Ridha Hamidi - 10/10/2023

A detailed configuration example that shows how to dual-home data center servers to Juniper leaf switches by using EZ-LAG, a simplified version on ESI-LAG made for customers looking for a smooth transition from Multi-Chassis LAG without having to immediately learn all the features and complexities of EVPN-VXLAN technology.

1.1 Introduction

Dual homing is an essential feature to provide connection redundancy for data center servers. Vendors have implemented multiple solutions to avoid single switch failure in data center fabrics, and to fulfill the need for redundant server connectivity, like Virtual-Chassis and MC-LAG/MLAG. These solutions are generally based on proprietary protocols and have variable levels of complexity for configuration and troubleshooting. In most recent years, and with the increasing adoption of EVPN-VXLAN as the founding technology for building modern and resilient data centers, ESI-LAG became the de-facto dual-homing solution between servers and Top-Of-Rack (TOR) switches. However, customers who are only looking for a better alternative to MC-LAG, adopting EVPN-VXLAN technology solely to fulfill the need for server dual-homing, is a steep transition and a hard decision to make, given the learning curve for this new technology.

1.2 Topology

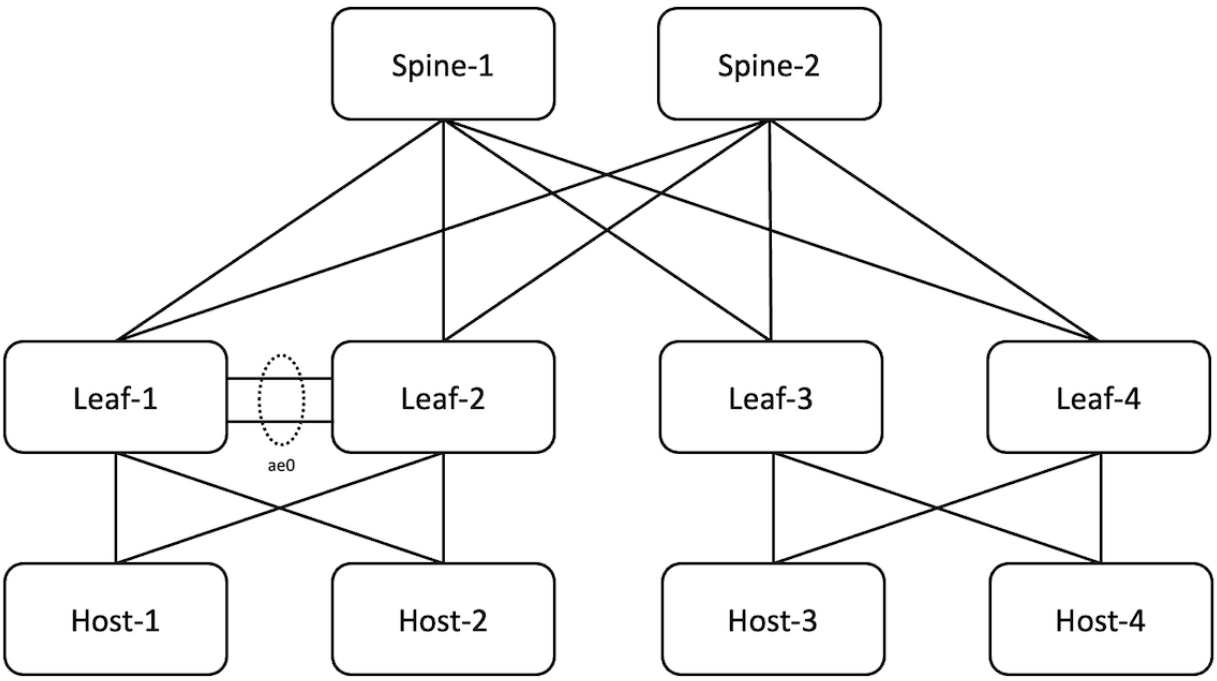


Figure 1: Reference Fabric Topology

- The fabric is preconfigured to provide basic underlay connectivity and all leafs and spines redistribute and receive all lookback addresses via the underlay protocol.

- In this post, we will use EBGP as the underlay protocol, but we will explain later that OSPF can be equally used.
- For the sake of comparing EZ-LAG to ESI-LAG and providing connectivity between all the hosts in the topology, the fabric is configured with BGP as an overlay protocol for family evpn, but only Leaf-3 and Leaf-4 are configured for EVPN-VXLAN. Therefore, Host-3 and Host-4 are dual homed to Leaf-3 and Leaf-4 with ESI-LAG and have overlay connectivity, but there is no connectivity between Host-1/Host-2 and Host-3/Host-4. The objective is to show how to achieve that connectivity by configuring EZ-LAG. Therefore, in the rest of this post, we will focus solely on the configuration of Leaf-1 and Leaf-2 for EZ-LAG.
- In this post, we will use EBGP as the overlay protocol, but we will explain later that IBGP can be equally used.

1.3 How Does EZ-LAG Work?

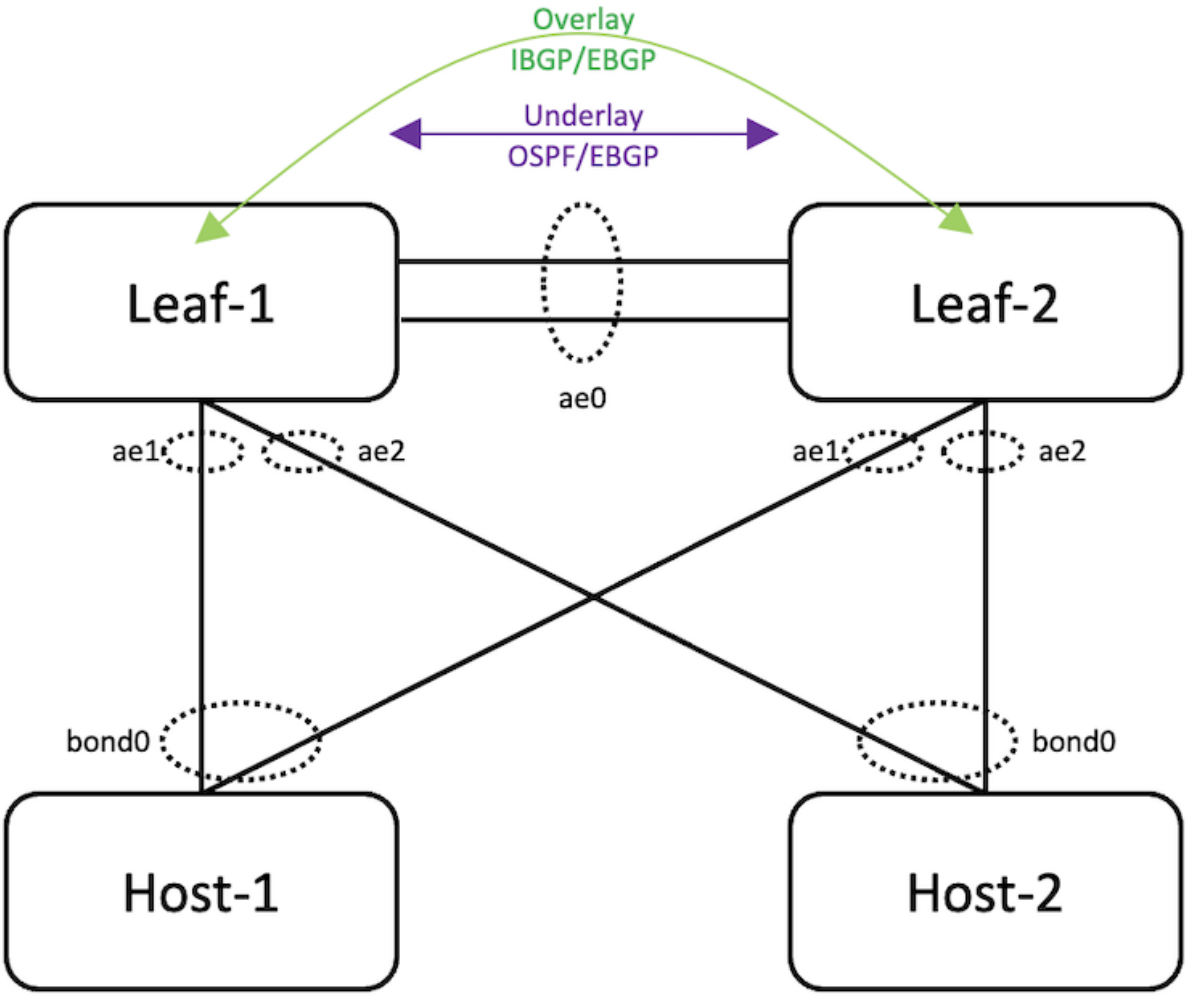


Figure 2: Reference Topology for EZ-LAG Configuration

- A new configuration hierarchy, called “evpn” under “services” stanza has been introduced specifically for EZ-LAG, where the user can specify a limited set of parameters to configure the feature.

- A Python script that the user can call via a commit script. The commit script takes the set of parameters specified by the user and derives all the necessary configuration to build the EZ-LAG. All configuration generated by the commit script will not be visible to the user, unless they explicitly specify “display commit-script”, that’s how the complexity of configuration is hidden from users who want to use the benefits of EZ-LAG feature, without carrying the complexity of its configuration.

```
show services evpn {
      defaults-override {
         no-aggregate-device-count-config;
         no-loop-detect-config;
         no-platform-defaults-config;
         no-policy-and-routing-options-config;
         no-storm-control-config;
         no-underlay-config; # hidden
      }
      global-parameters {
         start-aggregate-ethernet-index [ae-index];
         default-vlan-id [vlan-id];
         anycast-mac [mac-address];
         virtual-gateway {
            v4-mac [mac-address];
            v6-mac [mac-address];
         }
         virtual-gateway-mac [mac-address];
         no-irb-address-auto-derive;
      }
      device-attribute {
         peer-id [peer-id];
         loopback-subnet [subnet-address];
         system-id [system-id];
         peer-to-peer {
            peer-subnet {
               inet [ipv4-address];
               inet6 [ipv6-address];
               interface-name [interface-list];
            }
            overlay-connectivity {
               ibgp;
            }
            underlay-connectivity {
               ospf;
            }
         }
      }
      evpn-vxlan {
         dhcp-relay [relay-group] {
            relay-source [relay-source];
            dhcp-server-address [dhcp-server-address];
            vrf-instance [instance-id];
         }
         irb [irb-instance-name]
            use-anycast-address;
            subnet-address {
               inet [ipv4-subnet-address];
               inet6 [ipv6-subnet-address];
               virtual-gateway-v4-address [gateway-v4-address];
               virtual-gateway-v6-address [gateway-v6-address];
               no-irb-address-auto-derive;
            }
            no-dhcp-relay;
            vlan-id [vlan-id];
            instance {
               vrf-instance [instance-id];
               mac-vrf-instance [instance-id];
            }
            apply-config-groups [group-name];
         }
         server [server-name] {
            esi-lag-id [esi-lag-id];
            single-home-id [single-home-id];
            enable-pxe-boot;
            vlan-id-list [vlan-id-list];
            mac-vrf-instance [instance-id] vlan-id-list [vlan-id-list];
            interface [interface-name];
            apply-config-groups [group-name];
            no-lacp;
         }
```

```
72          }
73      }
74      traceoptions {
75          level {
76              all;
77              error;
78              info;
79              notice;
80              verbose;
81              warning;
82          }
83          flag {
84              all;
85              chassis;
86              forwarding-options;
87              input-configuration;
88              Interfaces;
89              policy-options;
90              Protocols;
91              routing-instance;
92              routing-options;
93      }
94  }
```

- “defaults-override” : all configuration knobs under this section are optional ; in this section, the user can specify all parameters they want to preserve, so the commit script will not override them. For example, if the user prefers to use a pre-configured ASN and specify “no-policy-and-routing-options-config” then they will need to configure everything under “ “routing-options” and “policy-options” hierarchies that are needed for EZ-LAG to function, like a Router-ID, a policy to export loopback addresses in the underlay, and a policy to load balance traffic exported to the forwarding table. Otherwise, an error will be thrown if you try to commit.
- “global-parameters” : all configuration knobs under this section are optional ; in this section, the user specifies the parameters that are common to both peers
- “device-attribute” : all configuration knobs under this section are mandatory, except “overlay-connectivity” ; this section contains parameters that are specific to the local peer. Parameters in this section must be carefully selected because must either match or be compatible with what is configured on the remote peer.
- “evpn-vxlan” : this section contains a mix of mandatory and optional parameters ; in this section, the user specifies the parameters needed to configure EVPN-VXLAN for EZ-LAG to function. This section requires a minimum understanding of how this feature works.

Table 1 below provides more details about the new configuration stanzas, attributes, their purpose, and if they are mandatory or optional:

Table 1: Parameters in Services EVPN Configuration Hierarchy

Object	Attribute	Description	Mandatory
default-override		All configuration parameters in this section are optional. In this section, the user specifies which configuration sections they want to configure manually and to prevent them from being over-written by auto-derived parameters.	No
	no-platform-defaults-config	Do not auto-derive platform specific configurations.	No
	no-aggregate-device-count-config	EZ-LAG needs at least two ae interfaces to function, ae0 for inter-chassis link, then one per dual-homed server.Device count must be configured otherwise the script will use the platform default, commonly 255	No

Object	Attribute	Description	Mandatory
	no-policy-and-routing-options-config	EZ-LAG script needs to configure autonomous-system and router-id under routing-options section, and per-packet load-balancing and export loopback under policy-options section. This knob id to allow the user to configure these manually.Note that if policies are configured manually, their names must be “pplb” and “EXPORT-LO0”	No
	no-loop-detect-config	This knob can be used to configure “protocols loop-detect” section manually. The auto-derived configuration of this section will be shown later in this post	No
	no-storm-control-config	This know lets the user configure storm-control manually under interfaces and forwarding-options sections. More details on this later in this post.	No
	no-underlay-config	This is a hidden knob that instructs the script not to generate an underlay configuration section.At the time of writing this post, this knob is not implemented	No
global-parameters		This section contains parameters that are common across peers	

Object	Attribute	Description	Mandatory
	start-aggregate-ethernet-index [ae-index]	By default, the script starts assigning dual-homed server interfaces starting from ae1. This knob allows to start the numbering from a different index to avoid potential conflicts	No
	default-vlan-id [vlan-id]	Specify the default VLAN ID. The default is system-default, generally 0	No
	anycast-mac [mac-address]	In case anycast is used for irb interfaces, this knob allows to configure the anycast MAC address.If virtual-gateway is used, this MAC address will be used as virtual-gateway-v4-mac	No
	no-irb-address-auto-derive	Do not auto-derive address and use the subnet address as is	No
virtual-gateway		You can configure either virtual-gateway-mac or virtual-gateway and then either v4-mac or v6-mac or both parameters	No
	v4-mac [mac-address]	The default is 00:00:5e:00:01:01	No
	v6-mac [mac-address]	The default is 00:00:5e:00:02:01	No
virtual-gateway-mac	[mac-address]	This applies both to v4 and v6 gateways	
device-attribute		This section contains parameters that are specific to local peer device	

Object	Attribute	Description	Mandatory
	peer-id [peer-id]	Local peer ID. It's important to choose this parameter carefully for both peers, because many other parameters are auto-derived by using this peer-idOne of the peers must have peer-id=1 and the other must be different than 1	Yes
	loopback-subnet [loopback-subnet]	lo0 prefix for deriving loopback address	Yes
	system-id [system-id]	System identifier in MAC address format ; this parameter must be identical on both peers, and will be used to auto-generate one system-id per ae interface, by incrementing by esi-lag-id for each ae interface.	Yes
peer-to-peer	peer-subnet [inet]	IPv4 address to be used for ae0 interface. Must be /31 subnetNOTE: either ipv4 or ipv6 is mandatory	Yes
	peer-subnet [inet6]	IPv6 address to be used for ae0 interface. Must be a /127 subnetNOTE: either ipv4 or ipv6 is mandatory	Yes
	peer-subnet interface [interface-name]	Peer to peer interface or list of interfaces between "[]" brackets	Yes
	overlay-connectivityibgp	Option to choose IBGP for the overlayNOTE: default is EBGP	No

Object	Attribute	Description	Mandatory
	underlay-connectivityospf	Option to choose OSPF for the underlayNOTE: default is EBGP	No
evpn-vxlan		Parameters for routing-instance and ESI-LAG configuration	
dhcp-relay		Dynamic Host Configuration Protocol relay configuration	
group-name		Group name to identify a Server/Relay group	No
	relay-source	Interface name for relay source	No
	dhcp-server-address [ip-address]	IP Address of one or more DHCP servers	No
	vrf-instance	Identifier for VRF instance (1..255)	No
irb-instance		Attribute specific to l3-interface/irb/etc	Yes
	use-anycast-address	Option to use anycast address instead of an irb gateway.If you configure this knob, you must also configure the anycast-mac address	No
	interface-name	Name of irb interface ; there could be multiple irb interfaces.	Yes
	subnet-address inet [ipv4-subnet-address]	Ipv4 address for irb gatewayNOTE: Either inet or inet6 is mandatory	See Note
	subnet-address inet6 [ipv6-subnet-address]	Ipv6 address for irb gatewayNOTE: Either inet or inet6 is mandatory	See Note
	subnet-address no-irb-address-auto-derive	Do not auto-derive address and use the subnet address as is	No

Object	Attribute	Description	Mandatory
	subnet-addressvirtual-gateway-v4-address	Virtual Gateway IPv4 address	No
	subnet-addressvirtual-gateway-v6-address	Virtual Gateway IPv6 address	No
	vlan-id [vlan-id]	IEEE 802.1q VLAN identifier for VLAN (1..4094)	Yes
	no-dhcp-relay	Exclude this irb from dhcp-relay	No
	apply-config-groups	Apply groups to the generated irb configuration	No
irb-instance instance			
	vrf-instance [instance-id]	VRF instance identifier (1..255)NOTE: default is 1	No
	mac-vrf-instance [instance-id]	MAC-VRF instance identifier (1..255)NOTE: default is 1	No
	apply-config-groups	Apply groups to the generated irb configuration	No
server			
	server-name	Server name ; each server must be assigned a server-name	Yes
	esi-lag-id	Assign an identifier for each server (1..1024)NOTE: Either esi-lag-id or single-home-id is mandatory	See Note
	single-home-id	Assign an identifier for a single home server (1..1024)NOTE: Either esi-lag-id or single-home-id is mandatory	See Note
	enable-pxe-boot	Enable PXE boot on this server	No

Object	Attribute	Description	Mandatory
	vlan-id-list [vlan-id-list]	List of VLANs for PEER to SERVERNOTE: With this config all VLANs will be in mac-vrf-instance 1NOTE: Either vlan-id-list or mac-vrf-instance vlan-id-list is mandatory	See Note
	interface [interface-name]	Interface or list of peer-to-server interfaces	Yes
	mac-vrf-instance [instance-id] vlan-id-list	List of VLANs for PEER to SERVER for mac-vrf-instanceNOTE: Either vlan-id-list or mac-vrf-instance vlan-id-list is mandatory	See Note
	apply-config-groups	Apply groups to the generated interface configuration	No
	no-lacp	Do not configure LACP for this server	No

Table 2: Default and Auto-Derived Parameters

Attribute	Description	Default ValueAuto-Derived Value
[ae-device-count]	number of AE devices in the system	255 or platform specific
[ae-peer-to-peer]	PEER to PEER aggregate interface name	ae0
[local-address]	PEER to PEER address for ae0 ; derived from[peer-subnet]parameter	Example:[peer-subnet] == 172.16.12.0/31if peer_id == 1:local_address = 172.16.12.0/31else:local_address = 172.16.12.1/31
[ae-peer-to-server]	PEER to SERVER aggregate interface name	for dual-homed server:ae[esi-lag-id]for single-homed server:ae[1024 + esi-lag-id]
[system-id]	LACP system id	[system-id] + [esi-lag-id]

Attribute	Description	Default ValueAuto-Derived Value
[loopback-address]	Loopback addressAlso, the router-id and the local-address of overlay BGP session	[loopback-subnet]Example:[loopback-subnet]:192.0.2.10/32[loopback-address]:192.0.2.10/32
[router-id]	Router identifier	[loopback-address]
[virtual-gateway-v4-mac]	Virtual gateway mac for v4	Default is 00:00:5e:00:01:01
[virtual-gateway-v6-mac]	Virtual gateway mac for v6	Default is 00:00:5e:00:02:01
[virtual-gateway-address]	Virtual gateway addressNOTE: this can be specified by the:set services evpn evpn-vxlan irb irb_10subnet-address inet [ipv4-subnet-address] virtual-gateway-v4-address [virtual-gateway-v4-address]set services evpn evpn-vxlan irb irb_10subnet-address inet6 [ipv6-subnet-address] virtual-gateway-v6-address [virtual-gateway-v6-address]	Rightmost digit of [subnet-address] set to 254For exampleIf [subnet-address] == 10.0.1.11/24:[virtual-gateway-address] = 10.0.1.254
[policy-statement]	Policy to load-balance “per-packet”, aka per flowset policy-options policy-statement pplb then load-balance per-packetset routing-options forwarding-table export pplb	pplb
[neighbor-address]	lo0 address of neighbor PEER	[loopback-subnet]+-1 for the leftmost IPv4 address digitFor example, [loopback-subnet] ==192.0.2.10/32If [peer-id] == 1:[neighbor-address] =193.0.2.10else[neighbor-address] =191.0.2.10
[underlay-interface]	Interface for OSPF	ae0.0
[routing-instance]	Routing instance name	__SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_[m vrf-instance]
[l3-interface]	Routing interface name	irb.[vlan-id]
[vlan-name]	Bridge domain name	SERVICES_EVPN_EVPN_VXLAN_VLAN_[vlan-id]
[vni]	VXLAN identifier	10000 + [vlan-id]

Attribute	Description	Default ValueAuto-Derived Value
[ae-peer-to-server-subunit]	PEER to SERVER aggregate interface subunit	ae[server-id].0
[route-distinguisher]	Route Distinguisher	[loopback-address]:[mac-vrf-instance]Example:[route-distinguisher]:192.0.2.10:1
[vrf-target]	VRF Target Extended Community	target:1:[mac-vrf-instance]
[ae-unit]	Unit for aggregated interface (server)	ae[esi-lag-id].[mac-vrf-instance]Example:ae1.1
[as_number]	Autonomous system number	65000 + [peer-id]
[local-as]	Local as number for underlay	if peer_id == 1:local_as = 65000 + 4else:local_as = 65000 + 3
[local-address]	Local address for underlay	[peer-subnet]+-1Example for [peer-subnet] = 172.16.12.0/31if peer_id == 1:local_address = 172.16.12.0else:local_address = 172.16.12.1
[neighbor]	Neighbor address for underlay	[peer-subnet]+-1Example for [peer-subnet] = 172.16.12.0/31if peer_id == 1:neighbor = 172.16.12.1else:neighbor = 172.16.12.0
[peer-as]	Peer as number for underlay	if peer_id == 1:peer_as = 65000 + 3else:peer_as = 65000 + 4
[local-address]	Local address for overlay	[loopback-address]
[neighbor]	Neighbor address for overlay	[loopback-subnet]+-1 for the leftmost IPv4 address digitFor example, [loopback-subnet] ==192.0.2.10/32if [peer-id] == 1:[neighbor-address] =193.0.2.10else[neighbor-address] =191.0.2.10
[peer-as]	Peer as number for overlay	if peer_id == 1:peer_as = 65000 + peer_id + 1else:peer_as = 65000 + peer_id - 1

In the setup we used for this post, these are the configured attributes:

	Leaf-1	Leaf-2
peer-id	1	2
loopback-subnet	111.1.1.10/32	112.1.1.10/32
system-id	00:14:15:16:17:10	00:14:15:16:17:10
peer-subnet	172.16.12.0/31	172.16.12.0/31
interface-name	xe-0/0/46	xe-0/0/46
irb instance interface-name	IRB_100	IRB_100
irb instance subnet-address	10.0.1.11/24	10.0.1.10/24
irb instancevlan-id	100	100
irb instance interface-name	IRB_200	IRB_200
irb instance subnet-address	10.0.2.11/24	10.0.2.10/24
irb instancevlan-id	200	200
server-name	SERVER_1	SERVER_1
server esi-lag-id	1	1
server vlan-id-list	100	100
interface-name	xe-0/0/45	xe-0/0/45
server-name	SERVER_2	SERVER_2
server esi-lag-id	2	2
irb instancevlan-id	200	200
interface-name	xe-0/0/44	xe-0/0/44

These are the auto derived attributes:

	Leaf-1	Leaf-2
ae-peer-to-peer	ae0	ae0
Local-address	172.16.12.0/31	172.16.12.1/31
ae-peer-to-server	ae1	ae1
LACP system-id	00:14:15:16:17:11	00:14:15:16:17:11
ae-unit	ae1.1	ae1.1
vlan members	100	100
VNI	10100	10100
l3-interface	Irb.100	Irb.100
virtual-gateway-address	10.0.1.254	10.0.1.254
ae-peer-to-server	ae2	ae2

	Leaf-1	Leaf-2
LACP system-id	00:14:15:16:17:12	00:14:15:16:17:12
ae-unit	ae2.1	ae2.1
vlan members	200	200
VNI	10200	10200
l3-interface	Irb.200	Irb.200
virtual-gateway-address	10.0.2.254	10.0.2.254
MAC-VRF instance name	__SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1	__SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1
Loopback address	111.1.1.10/32	112.1.1.10/32
router-id	111.1.1.10	112.1.1.10
route-distinguisher	111.1.1.10:1	112.1.1.10:1
vrf-target	target:1:1	target:1:1
autonomous-system	65001	65002
Underlay BGP local-address	172.16.12.0	172.16.12.1
Underlay BGP local-as	65004	65003
Underlay BGP neighbor	172.16.12.1	172.16.12.0
Underlay peer-as	65003	65004
Overlay BGP local-address	111.1.1.10	112.1.1.10
Overlay BGP neighbor	112.1.1.10	111.1.1.10
Overlay BGP peer-as	65002	65001

1.4 Basic EZ-LAG Configuration

1.4.1 Peer 1 Configuration

Peer 1 Base Configuration

```
set version 23.2R1.13
set system host-name qfx1
set system root-authentication encrypted-password "[secret data]"
set system services ssh root-login allow
set system syslog user * any emergency
set system syslog file interactive-commands interactive-commands any
set system syslog file messages any notice
set system syslog file messages authorization info
set interfaces em0 unit 0 family inet address 192.168.1.101/24
set forwarding-options evpn-vxlan shared-tunnels
set routing-options static route 0.0.0.0/0 next-hop 192.168.1.1
set protocols lldp port-id-subtype interface-name
set protocols lldp interface all
```

Peer 1 EZ-LAG Configuration

```
set system scripts commit allow-transients
set system scripts commit file services_evpn_commit_script.py
set services evpn device-attribute peer-id 1
```

```
set services evpn device-attribute loopback-subnet 111.1.1.10/32
set services evpn device-attribute system-id 00:14:15:16:17:11
set services evpn device-attribute peer-to-peer peer-subnet inet 172.16.12.0/31
set services evpn device-attribute peer-to-peer peer-subnet interface-name xe-0/0/46
set services evpn evpn-vxlan irb IRB_100 vlan-id 100
set services evpn evpn-vxlan irb IRB_100 subnet-address inet 10.0.1.11/24
set services evpn evpn-vxlan irb IRB_200 vlan-id 200
set services evpn evpn-vxlan irb IRB_200 subnet-address inet 10.0.2.11/24
set services evpn evpn-vxlan server SERVER_1 esi-lag-id 1
set services evpn evpn-vxlan server SERVER_1 vlan-id-list 100
set services evpn evpn-vxlan server SERVER_1 interface xe-0/0/45
set services evpn evpn-vxlan server SERVER_2 esi-lag-id 2
set services evpn evpn-vxlan server SERVER_2 vlan-id-list 200
set services evpn evpn-vxlan server SERVER_2 interface xe-0/0/44
```

1.4.2 Peer 2 Configuration

Peer 2 Base Configuration

```
set version 23.2R1.13
set system host-name qfx2
set system root-authentication encrypted-password "[secret data]"
set system services ssh root-login allow
set system syslog user * any emergency
set system syslog file interactive-commands interactive-commands any
set system syslog file messages any notice
set system syslog file messages authorization info
set interfaces em0 unit 0 family inet address 192.168.1.102/24
set forwarding-options evpn-vxlan shared-tunnels
set routing-options static route 0.0.0.0/0 next-hop 192.168.1.1
set protocols lldp port-id-subtype interface-name
set protocols lldp interface all
```

Peer 2 EZ-LAG Configuration

```
set system scripts commit allow-transients
set system scripts commit file services_evpn_commit_script.py
set services evpn device-attribute peer-id 7
set services evpn device-attribute loopback-subnet 111.1.1.10/32
set services evpn device-attribute system-id 00:14:15:16:17:10
set services evpn device-attribute peer-to-peer peer-subnet inet 172.16.12.0/31
set services evpn device-attribute peer-to-peer peer-subnet interface-name xe-0/0/46
set services evpn evpn-vxlan irb IRB_100 vlan-id 100
set services evpn evpn-vxlan irb IRB_100 subnet-address inet 10.0.1.10/24
set services evpn evpn-vxlan irb IRB_200 vlan-id 200
set services evpn evpn-vxlan irb IRB_200 subnet-address inet 10.0.2.10/24
set services evpn evpn-vxlan server SERVER_1 esi-lag-id 1
set services evpn evpn-vxlan server SERVER_1 vlan-id-list 100
set services evpn evpn-vxlan server SERVER_1 vlan-id-list 200
set services evpn evpn-vxlan server SERVER_1 interface xe-0/0/45
set services evpn evpn-vxlan server SERVER_2 esi-lag-id 2
set services evpn evpn-vxlan server SERVER_2 vlan-id-list 100
set services evpn evpn-vxlan server SERVER_2 vlan-id-list 200
set services evpn evpn-vxlan server SERVER_2 interface xe-0/0/44
```

1.4.3 Host-1 Configuration Files (Centos 7)

/etc/sysconfig/network-scripts/ifcfg-ens2f0

```
NAME=ens2f0
UUID=4f8772aa-1c36-44d8-a1d5-8da3d7921eec
DEVICE=ens2f0
TYPE=Ethernet
BOOTPROTO=none
ONBOOT=yes
NM_CONTROLLED=no
SLAVE=yes
MASTER=bond0
```

/etc/sysconfig/network-scripts/ifcfg-ens2f1

```
NAME=ens2f1
DEVICE=ens2f1
TYPE=Ethernet
UUID=9b89bf2d-2d26-49e8-b1d1-c532816e1dcb
BOOTPROTO=none
ONBOOT=no
NM_CONTROLLED=no
SLAVE=yes
MASTER=bond0
```

/etc/sysconfig/network-scripts/ifcfg-bond0

```
DEVICE=bond0
TYPE=Bond
ONBOOT=yes
BONDING_OPTS="mode=802.3ad miimon=100 lacp_rate=fast"
NM_CONTROLLED=no
MTU="9000"
```

/etc/sysconfig/network-scripts/ifcfg-bond0.100

```
DEVICE=bond0.100
ONPARENT=yes
BOOTPROTO=none
DEFROUTE=yes
IPADDR=10.0.1.111
PREFIX=24
VLAN=yes
NM_CONTROLLED=no
ZONE=trusted
```

/etc/sysconfig/network-scripts/route-bond0.100

```
10.0.2.0/24 via 10.0.1.254
10.0.3.0/24 via 10.0.1.254
10.0.4.0/24 via 10.0.1.254
```

1.4.4 Host-2 Configuration (Ubuntu 20.04)

```
network:
   version: 2
   renderer: networkd
   ethernets:
      ens2f0:
        match:
          macaddress: a0:42:3f:3a:f7:76
        set-name: ens2f0
      ens2f1:
        match:
          macaddress: a0:42:3f:3a:f7:77
        set-name: ens2f1
   bonds:
      bond0:
        dhcp4: false
        dhcp6: false
        interfaces: [ ens2f0, ens2f1 ]
        parameters:
          mode: 802.3ad
          mii-monitor-interval: 100
   vlans:
      bond0.200:
        id: 200
        link: bond0
        addresses : [10.0.2.2/24]
        routes:
        - to: 10.0.1.0/24
          via: 10.0.2.1
        - to: 10.0.3.0/24
          via: 10.0.2.1
        - to: 10.0.4.0/24
          via: 10.0.2.1
```

1.5 Verifications

Let's now look at the details of the configuration that has been generated by the commit script and verify that the setup is working as expected.

As explained at the beginning of this post, since the whole objective of this feature is to not expose the complexity of EVPN-VXLAN configuration to the user, we chose to implement it by using a commit script. The curious user can refer to the document listed in the "Useful Links" section to learn more about how commit scripts work on a Junos device. For the purpose of this post, we can simply say that a commit script is a tool that allows the user to verify at commit time that the configuration complies with certain user defined rules, and generate warnings, logs, or even change the configuration itself. It is the latter option that is used to implement EZ-LAG feature. In this case, when the user commits the configuration above, the commit script parses "services evpn" section of the configuration and generates other configuration sections that are not exposed by default in the configuration displayed with "show configuration" command. However, the user can always use "display commit-scripts" to see the entire configuration, the original one and the commit-script generated one, all merged into one configuration, with certain markers for easier reading.

The excerpt below shows the output of this special "show" command on Leaf-1 after committing the configuration listed above.

We can see that the commit script auto generated CLI configuration commands under interfaces, forwarding-options, policy-options, routing-instances, routing-options, and protocols stanzas.

```
{master:0}
root@qfx1] show configuration | display commit-scripts | display set
set version 23.2R1.13
set system host-name qfx1
set system root-authentication encrypted-password "$6$8xY0iVur$w8.f59mL/
    q7w4VtO4KZCCSape0vDokgktMxfJ2Z.MY.1pBjvVR5wPlhYiG.3sg31UsxjHnnWjHezBJ6kAQYnd0"
set system scripts commit allow-transients
set system scripts commit file services_evpn_commit_script.py
set system services ssh root-login allow
set system syslog user * any emergency
set system syslog file interactive-commands interactive-commands any
set system syslog file messages any notice
set system syslog file messages authorization info
set chassis aggregated-devices ethernet device-count 255
set chassis network-services enhanced-ip
set services evpn device-attribute peer-id 1
set services evpn device-attribute loopback-subnet 111.1.1.10/32
set services evpn device-attribute system-id 00:01:02:03:04:05
set services evpn device-attribute peer-to-peer peer-subnet inet 172.16.12.0/31
set services evpn device-attribute peer-to-peer peer-subnet interface-name xe-0/0/46
set services evpn evpn-vxlan irb IRB_100 vlan-id 100
set services evpn evpn-vxlan irb IRB_100 subnet-address inet 10.0.1.11/24
set services evpn evpn-vxlan irb IRB_200 vlan-id 200
set services evpn evpn-vxlan irb IRB_200 subnet-address inet 10.0.2.11/24
set services evpn evpn-vxlan server SERVER_1 esi-lag-id 1
set services evpn evpn-vxlan server SERVER_1 vlan-id-list 100
set services evpn evpn-vxlan server SERVER_1 interface xe-0/0/45
set services evpn evpn-vxlan server SERVER_2 esi-lag-id 2
set services evpn evpn-vxlan server SERVER_2 vlan-id-list 200
set services evpn evpn-vxlan server SERVER_2 interface xe-0/0/44
set interfaces xe-0/0/44 ether-options ethernet-switch-profile storm-control
    __SERVICES_EVPN_EVPN_VXLAN_STORM_CONTROL
set interfaces xe-0/0/44 ether-options 802.3ad ae2
set interfaces xe-0/0/45 ether-options ethernet-switch-profile storm-control
    __SERVICES_EVPN_EVPN_VXLAN_STORM_CONTROL
set interfaces xe-0/0/45 ether-options 802.3ad ae1
set interfaces xe-0/0/46 ether-options 802.3ad ae0
set interfaces ae0 aggregated-ether-options lacp active
set interfaces ae0 unit 0 family inet address 172.16.12.0/31
set interfaces ae1 vlan-tagging
set interfaces ae1 encapsulation flexible-ethernet-services
set interfaces ae1 esi auto-derive type-1-lacp
set interfaces ae1 esi all-active
set interfaces ae1 aggregated-ether-options lacp active
```

```
set interfaces ae1 aggregated-ether-options lacp system-id 00:01:02:03:04:06
set interfaces ae1 unit 1 family ethernet-switching interface-mode trunk
set interfaces ae1 unit 1 family ethernet-switching vlan members 100
set interfaces ae2 vlan-tagging
set interfaces ae2 encapsulation flexible-ethernet-services
set interfaces ae2 esi auto-derive type-1-lacp
set interfaces ae2 esi all-active
set interfaces ae2 aggregated-ether-options lacp active
set interfaces ae2 aggregated-ether-options lacp system-id 00:01:02:03:04:07
set interfaces ae2 unit 1 family ethernet-switching interface-mode trunk
set interfaces ae2 unit 1 family ethernet-switching vlan members 200
set interfaces em0 unit 0 family inet address 10.87.11.81/25
set interfaces irb unit 100 virtual-gateway-accept-data
set interfaces irb unit 100 family inet address 10.0.1.11/24 virtual-gateway-address 10.0.1.254
set interfaces irb unit 100 virtual-gateway-v4-mac 00:00:5e:00:01:01
set interfaces irb unit 200 virtual-gateway-accept-data
set interfaces irb unit 200 family inet address 10.0.2.11/24 virtual-gateway-address 10.0.2.254
set interfaces irb unit 200 virtual-gateway-v4-mac 00:00:5e:00:01:01
set interfaces lo0 unit 0 family inet address 111.1.1.10/32 primary
set interfaces lo0 unit 0 family inet address 111.1.1.10/32 preferred
set forwarding-options storm-control-profiles __SERVICES_EVPN_EVPN_VXLAN_STORM_CONTROL all
    bandwidth-percentage 1
set forwarding-options evpn-vxlan shared-tunnels
set forwarding-options vxlan-routing next-hop 45056
set forwarding-options vxlan-routing interface-num 8192
set forwarding-options vxlan-routing overlay-ecmp
set policy-options policy-statement EXPORT-LO0 term LOOPBACK from interface lo0.0
set policy-options policy-statement EXPORT-LO0 term LOOPBACK then accept
set policy-options policy-statement EXPORT-LO0 term REJECT then reject
set policy-options policy-statement pplb then load-balance per-packet
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 instance-type mac-vrf
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 protocols evpn encapsulation vxlan
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 protocols evpn default-gateway no-
    gateway-community
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 vtep-source-interface lo0.0
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 service-type vlan-aware
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 interface ae1.1
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 interface ae2.1
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 route-distinguisher 111.1.1.10:1
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 vrf-target target:1:1
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 vlans
    SERVICES_EVPN_EVPN_VXLAN_VLAN_100 vlan-id 100
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 vlans
    SERVICES_EVPN_EVPN_VXLAN_VLAN_100 l3-interface irb.100
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 vlans
    SERVICES_EVPN_EVPN_VXLAN_VLAN_100 vxlan vni 10100
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 vlans
    SERVICES_EVPN_EVPN_VXLAN_VLAN_200 vlan-id 200
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 vlans
    SERVICES_EVPN_EVPN_VXLAN_VLAN_200 l3-interface irb.200
set routing-instances __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1 vlans
    SERVICES_EVPN_EVPN_VXLAN_VLAN_200 vxlan vni 10200
set routing-options router-id 111.1.1.10
set routing-options autonomous-system 65001
set routing-options static route 0.0.0.0/0 next-hop 10.87.11.126
set routing-options forwarding-table export pplb
set protocols bgp group __SERVICES_EVPN_EVPN_VXLAN_EBGP_UNDERLAY type external
set protocols bgp group __SERVICES_EVPN_EVPN_VXLAN_EBGP_UNDERLAY local-address 172.16.12.0
set protocols bgp group __SERVICES_EVPN_EVPN_VXLAN_EBGP_UNDERLAY export EXPORT-LO0
set protocols bgp group __SERVICES_EVPN_EVPN_VXLAN_EBGP_UNDERLAY local-as 65004
set protocols bgp group __SERVICES_EVPN_EVPN_VXLAN_EBGP_UNDERLAY multipath multiple-as
set protocols bgp group __SERVICES_EVPN_EVPN_VXLAN_EBGP_UNDERLAY neighbor 172.16.12.1 peer-as
    65003
set protocols bgp group __SERVICES_EVPN_EVPN_VXLAN_EBGP_OVERERLAY type external
set protocols bgp group __SERVICES_EVPN_EVPN_VXLAN_EBGP_OVERERLAY multihop ttl 2
set protocols bgp group __SERVICES_EVPN_EVPN_VXLAN_EBGP_OVERERLAY multihop no-nexthop-change
set protocols bgp group __SERVICES_EVPN_EVPN_VXLAN_EBGP_OVERERLAY local-address 111.1.1.10
set protocols bgp group __SERVICES_EVPN_EVPN_VXLAN_EBGP_OVERERLAY family evpn signaling
set protocols bgp group __SERVICES_EVPN_EVPN_VXLAN_EBGP_OVERERLAY neighbor 112.1.1.10 peer-as
    65002
set protocols evpn no-core-isolation
set protocols lldp port-id-subtype interface-name
set protocols lldp interface all
set protocols loop-detect enhanced interface ae1 vlan-id 100
set protocols loop-detect enhanced interface ae1 loop-detect-action interface-down
set protocols loop-detect enhanced interface ae1 transmit-interval 1s
set protocols loop-detect enhanced interface ae1 revert-interval 60
set protocols loop-detect enhanced interface ae2 vlan-id 200
set protocols loop-detect enhanced interface ae2 loop-detect-action interface-down
set protocols loop-detect enhanced interface ae2 transmit-interval 1s
set protocols loop-detect enhanced interface ae2 revert-interval 60
```

You see the benefits of EZ-LAG feature by comparing the length of the original config to that of the auto-generated one and understand the time saving to configure it and having less chances to make configuration mistakes at the same time.

Notice the highlighted system-ids for both ae interfaces that are autogenerated and must match on the two peers.

Let's now verify that the setup is working as expected by running some operational CLI commands.

Leaf-1

```
{master:0}
root@qfx1] show interfaces terse
Interface Admin Link Proto Local Remote

[snip]

xe-0/0/44 up up
xe-0/0/44.1 up up aenet --] ae2.1
xe-0/0/44.32767 up up aenet --] ae2.32767
xe-0/0/45 up up
xe-0/0/45.1 up up aenet --] ae1.1
xe-0/0/45.32767 up up aenet --] ae1.32767
xe-0/0/46 up up
xe-0/0/46.0 up up aenet --] ae0.0

[snip]

ae0 up up
ae0.0 up up inet 172.16.12.0/31
ae1 up up
ae1.1 up up eth-switch
ae1.32767 up up
ae2 up up
ae2.1 up up eth-switch
ae2.32767 up up

[snip]

irb up up
irb.100 up up inet 10.0.1.11/24
                                  10.0.1.254/24
irb.200 up up inet 10.0.2.11/24
                                  10.0.2.254/24

[snip]

lo0.0 up up inet 111.1.1.10 --] 0/0
lo0.16384 up up inet 127.0.0.1 --] 0/0
lo0.16385 up up inet

[snip]
```

Leaf-2

```
{master:0}
root@qfx2] show interfaces terse
Interface Admin Link Proto Local Remote
gr-0/0/0 up up
pfe-0/0/0 up up
pfe-0/0/0.16383 up up inet
                       inet6
pfh-0/0/0 up up
pfh-0/0/0.16383 up up inet
pfh-0/0/0.16384 up up inet
xe-0/0/44 up up
xe-0/0/44.1 up up aenet --] ae2.1
xe-0/0/44.32767 up up aenet --] ae2.32767
xe-0/0/45 up up
xe-0/0/45.1 up up aenet --] ae1.1
xe-0/0/45.32767 up up aenet --] ae1.32767
xe-0/0/46 up up
xe-0/0/46.0 up up aenet --] ae0.0
```

```
et-0/0/48 up up
et-0/0/48.16386 up up
et-0/0/50 up up
et-0/0/50.16386 up up
ae0 up up
ae0.0 up up inet 172.16.12.1/31
ae1 up up
ae1.1 up up eth-switch
ae1.32767 up up
ae2 up up
ae2.1 up up eth-switch
ae2.32767 up up
bme0 up up
bme0.0 up up inet 128.0.0.1/2
                                  128.0.0.4/2
                                  128.0.0.63/2
cbp0 up up
dsc up up
em0 up up
em0.0 up up inet 10.87.11.82/25
em1 up down
em2 up up
em2.32768 up up inet 192.168.1.2/24
em3 up up
esi up up
fti0 up up
gre up up
ipip up up
irb up up
irb.100 up up inet 10.0.1.11/24
                                  10.0.1.254/24
irb.200 up up inet 10.0.2.11/24
                                  10.0.2.254/24
jsrv up up
jsrv.1 up up inet 128.0.0.127/2
lo0 up up
lo0.0 up up inet 112.1.1.10 --] 0/0
lo0.16384 up up inet 127.0.0.1 --] 0/0
lo0.16385 up up inet
lsi up up
mtun up up
pimd up up
pime up up
pip0 up up
tap up up
vme up down
vtep up up
vtep.32768 up up
vtep.32769 up up eth-switch
vtep.32771 up up

{master:0}
root@qfx2]
```

The above excerpt shows that the script created interfaces ae0.0, ae1.1, ae2.1, irb.100, irb.200, and lo0.0 and assigned the proper addresses.

It's also important to verify that ae1 and ae2 interfaces are "up up" on both leafs, because ESI is configured in all-active mode.

Let's first look at LACP status and statistics. It is important that Host-1 sees Leaf-1 and Leaf-2 as the same LACP system, therefore it does not know that it is dual-homed to two separate leafs. This is verified below because the System ID is identical for the "Actor" role on both leafs.

Leaf-1

```
{master:0}
root@qfx1] show lacp interfaces ae1 extensive
Aggregated interface: ae1
   LACP state: Role Exp Def Dist Col Syn Aggr Timeout Activity
     xe-0/0/45 Actor No No Yes Yes Yes Yes Fast Active
     xe-0/0/45 Partner No No Yes Yes Yes Yes Fast Active
   LACP protocol: Receive State Transmit State Mux State
     xe-0/0/45 Current Fast periodic Collecting distributing
   LACP info: Role System System Port Port Port
```

```
                      priority identifier priority number key
     xe-0/0/45 Actor 127 00:01:02:03:04:06 127 2 2
     xe-0/0/45 Partner 65535 90:e2:ba:d9:30:64 255 1 15

{master:0}
root@qfx1]
```

Leaf-2

```
{master:0}
root@qfx2] show lacp interfaces ae1 extensive
Aggregated interface: ae1
   LACP state: Role Exp Def Dist Col Syn Aggr Timeout Activity
     xe-0/0/45 Actor No No Yes Yes Yes Yes Fast Active
     xe-0/0/45 Partner No No Yes Yes Yes Yes Fast Active
   LACP protocol: Receive State Transmit State Mux State
     xe-0/0/45 Current Fast periodic Collecting distributing
   LACP info: Role System System Port Port Port
                      priority identifier priority number key
     xe-0/0/45 Actor 127 00:01:02:03:04:06 127 2 2
     xe-0/0/45 Partner 65535 90:e2:ba:d9:30:64 255 2 15

{master:0}
root@qfx2]
```

Let's now have a closer look at one of the AE interfaces.

Leaf-1

```
{master:0}
root@qfx1] show interfaces ae1 extensive
Physical interface: ae1, Enabled, Physical link is Up
  Interface index: 692, SNMP ifIndex: 522, Generation: 183
  Link-level type: Flexible-Ethernet, MTU: 1518, Speed: 10Gbps, BPDU Error: None, Ethernet-
      Switching Error: None, MAC-REWRITE Error: None, Loopback: Disabled, Source filtering:
      Disabled,
  Flow control: Disabled, Minimum links needed: 1, Minimum bandwidth needed: 1bps
  Device flags : Present Running
  Interface flags: SNMP-Traps Internal: 0x4000
  Current address: 80:7f:f8:d2:d0:29, Hardware address: 80:7f:f8:d2:d0:29
  Ethernet segment value: 01:90:e2:ba:d9:30:64:00:0f:00, Mode: all-active
  Last flapped : 2023-08-28 09:28:48 UTC (02:31:09 ago)
  Statistics last cleared: Never
  Traffic statistics:
   Input bytes : 1297388 1256 bps
   Output bytes : 1332988 1256 bps
   Input packets: 9983 1 pps
   Output packets: 9957 1 pps
   IPv6 transit statistics:
   Input bytes : 0
   Output bytes : 0
   Input packets: 0
   Output packets: 0
  Input errors:
    Errors: 0, Drops: 0, Framing errors: 0, Runts: 0, Giants: 0, Policed discards: 0, Resource
        errors: 0
  Output errors:
    Carrier transitions: 1, Errors: 0, Drops: 0, MTU errors: 0, Resource errors: 0
  Egress queues: 10 supported, 5 in use
  Queue counters: Queued packets Transmitted packets Dropped packets
    0 566 566 0
    3 0 0 0
    4 0 0 0
    7 9386 9386 0
    8 9 9 0
  Queue number: Mapped forwarding classes
    0 best-effort
    3 fcoe
    4 no-loss
    7 network-control
    8 mcast

  Logical interface ae1.1 (Index 878) (SNMP ifIndex 537) (HW Token 1) (Generation 339)
    Flags: Up SNMP-Traps 0x20024000 Encapsulation: Ethernet-Bridge
    Statistics Packets pps Bytes bps
    Bundle:
       Input : 0 0 0 0
       Output: 3 0 138 0
```

```
    Adaptive Statistics:
       Adaptive Adjusts: 0
       Adaptive Scans : 0
       Adaptive Updates: 0
    Link:
      xe-0/0/45.1
       Input : 0 0 0 0
       Output: 3 0 138 0

    Aggregate member links: 1

    Marker Statistics: Marker Rx Resp Tx Unknown Rx Illegal Rx
      xe-0/0/45.1 0 0 0 0
    Protocol eth-switch, MTU: 1518, Generation: 305, Route table: 8, Mesh Group: __all_ces__,
        EVPN multi-homed status: Forwarding, EVPN multi-homed ESI Split Horizon Label: 0, Next-
        hop: 1804,
    vpls-status: up
    Local Bias Logical Interface Name: vtep.32770, Index: 881, VXLAN Endpoint Address: 112.1.1.10
      Flags: Trunk-Mode, 0x4000000

  Logical interface ae1.32767 (Index 879) (SNMP ifIndex 529) (Generation 340)
    Flags: Up SNMP-Traps 0x4004000 VLAN-Tag [ 0x0000.0 ] Encapsulation: ENET2
    Statistics Packets pps Bytes bps
    Bundle:
       Input : 0 0 0 0
       Output: 0 0 0 0
    Adaptive Statistics:
       Adaptive Adjusts: 0
       Adaptive Scans : 0
       Adaptive Updates: 0
    Link:
      xe-0/0/45.32767
       Input : 0 0 0 0
       Output: 329 0 134873 0
    LACP info: Role System System Port Port Port
                       priority identifier priority number key
      xe-0/0/45.32767 Actor 127 00:01:02:03:04:06 127 2 2
      xe-0/0/45.32767 Partner 65535 90:e2:ba:d9:30:64 255 1 15
    LACP Statistics: LACP Rx LACP Tx Unknown Rx Illegal Rx
      xe-0/0/45.32767 9048 9007 0 0
    Marker Statistics: Marker Rx Resp Tx Unknown Rx Illegal Rx
      xe-0/0/45.32767 0 0 0 0

{master:0}
root@qfx1]
```

EZ-LAG sets LACP auto-derive mode to “type-1-lacp” and, in this mode, the ESI is the system ID and the administrative key on the remote CE device in the LACP link (partner). In our case, the remote CE device is Host-1

- 01 - ESI Type-1
- 90:e2:ba:d9:30:64 - MAC address of the bond on Host-1
- 00:0f - admin key on Host-1
- 00 - the last octet is always 0

These information can be easily verified on Host-1

```
[root@5a7s38 ~]# ifconfig bond0 | grep ether
      ether 90:e2:ba:d9:30:64 txqueuelen 1000 (Ethernet)
[root@5a7s38 ~]#
[root@5a7s38 ~]# tcpdump -i ens2f0 -vvv
tcpdump: listening on ens2f0, link-type EN10MB (Ethernet), capture size 262144 bytes
19:53:42.884097 LACPv1, length 110
      Actor Information TLV (0x01), length 20
        System 90:e2:ba:d9:30:64 (oui Unknown), System Priority 65535, Key 15, Port 1, Port
            Priority 255
        State Flags [Activity, Timeout, Aggregation, Synchronization, Collecting, Distributing]
        0x0000: ffff 90e2 bad9 3064 000f 00ff 0001 3f00
        0x0010: 0000
      Partner Information TLV (0x02), length 20
        System 00:01:02:03:04:06 (oui Unknown), System Priority 127, Key 2, Port 2, Port
            Priority 127
        State Flags [Activity, Timeout, Aggregation, Synchronization, Collecting, Distributing]
        0x0000: 007f 0001 0203 0406 0002 007f 0002 3f00
```

```
        0x0010: 0000
      Collector Information TLV (0x03), length 16
        Max Delay 0
        0x0000: 0000 0000 0000 0000 0000 0000 0000
      Terminator TLV (0x00), length 0
```

Verify that ESI is identical on Leaf-2

```
{master:0}
root@qfx2] show interfaces ae1 extensive | match segment
  Ethernet segment value: 01:90:e2:ba:d9:30:64:00:0f:00, Mode: all-active

{master:0}
root@qfx2]
```

It is important to verify that for each ae interface, only one leaf is forwarding BUM traffic to the server and the other leaf is blocking it.

```
{master:0}
root@qfx2] show interfaces ae1 extensive | match multi-homed
   Protocol eth-switch, MTU: 1518, Generation: 301, Route table: 8, Mesh Group: __all_ces__,
        EVPN multi-homed status: Blocking BUM Traffic to ESI, EVPN multi-homed ESI Split Horizon
        Label: 0,

{master:0}
root@qfx2]
```

In our case, we can see that EVPN multi-homed status on Leaf-1 is “Forwarding” and that on Leaf-2, the status is “Blocking BUM traffic to ESI”.

From BGP perspective, we need to have two BGP sessions, one for the underlay and another for the overlay. Here both are “Established” and Leaf-1 received 12 EVPN routes.

```
{master:0}
root@qfx1] show bgp summary
Threading mode: BGP I/O
Default eBGP mode: advertise - accept, receive - accept
Groups: 2 Peers: 2 Down peers: 0
Table Tot Paths Act Paths Suppressed History Damp State Pending
bgp.evpn.0
                  12 12 0 0 0 0
inet.0
                   1 1 0 0 0 0
Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
112.1.1.10 65002 130 131 0 0 55:16 Establ
  bgp.evpn.0: 12/12/12/0
  __SERVICES_EVPN_EVPN_VXLAN_MAC_VRF_1.evpn.0: 12/12/12/0
  __default_evpn__.evpn.0: 0/0/0/0
172.16.12.1 65003 127 125 0 0 55:19 Establ
  inet.0: 1/1/1/0

{master:0}
root@qfx1]
```

Now, it’s important to check EVPN routes that have been exchanged between the two peers, after verifying reachability between Host-1 and Host-2.

In the context of this post, we’re more particularly interested in Type 1 EVPN routes that are used for per Ethernet segment auto-discovery to advertise EVPN multi-homing mode.

```
{master:0}
root@qfx1] show route table bgp.evpn.0 protocol bgp

bgp.evpn.0: 44 destinations, 44 routes (44 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

1:112.1.1.10:0::0190e2bad1e230000f00::FFFF:FFFF/192 AD/ESI
               *[BGP/170] 02:25:52, localpref 100, from 112.1.1.10
                 AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0
1:112.1.1.10:0::0190e2bad93064000f00::FFFF:FFFF/192 AD/ESI
```

```
                *[BGP/170] 02:25:53, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0
1:112.1.1.1.10:0::050000fdea0000277400::FFFF:FFFF/192 AD/ESI
                *[BGP/170] 04:09:41, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0
1:112.1.1.1.10:0::050000fdea000027d800::FFFF:FFFF/192 AD/ESI
                *[BGP/170] 04:09:41, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0
1:112.1.1.1.10:1::0190e2bad1e230000f00::0/192 AD/EVI
                *[BGP/170] 02:25:53, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0
1:112.1.1.1.10:1::0190e2bad93064000f00::0/192 AD/EVI
                *[BGP/170] 02:25:54, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0
2:112.1.1.1.10:1::10100::00:00:5e:00:01:01/304 MAC/IP
                *[BGP/170] 04:09:41, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0, Push 631
2:112.1.1.1.10:1::10100::80:7f:f8:d2:02:60/304 MAC/IP
                *[BGP/170] 04:09:41, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0, Push 631
2:112.1.1.1.10:1::10100::90:e2:ba:d9:30:64/304 MAC/IP
                *[BGP/170] 01:34:16, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0, Push 631
2:112.1.1.1.10:1::10200::00:00:5e:00:01:01/304 MAC/IP
                *[BGP/170] 04:09:41, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0, Push 637
2:112.1.1.1.10:1::10200::80:7f:f8:d2:02:60/304 MAC/IP
                *[BGP/170] 04:09:41, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0, Push 637
2:112.1.1.1.10:1::10200::90:e2:ba:d1:e2:30/304 MAC/IP
                *[BGP/170] 01:34:16, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0, Push 637
2:112.1.1.1.10:1::10100::00:00:5e:00:01:01::10.0.1.254/304 MAC/IP
                *[BGP/170] 04:09:41, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0, Push 631
2:112.1.1.1.10:1::10100::80:7f:f8:d2:02:60::10.0.1.11/304 MAC/IP
                *[BGP/170] 04:09:41, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0, Push 631
2:112.1.1.1.10:1::10100::90:e2:ba:d9:30:64::10.0.1.111/304 MAC/IP
                *[BGP/170] 00:19:07, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0, Push 631
2:112.1.1.1.10:1::10200::00:00:5e:00:01:01::10.0.2.254/304 MAC/IP
                *[BGP/170] 04:09:41, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0, Push 637
2:112.1.1.1.10:1::10200::80:7f:f8:d2:02:60::10.0.2.11/304 MAC/IP
                *[BGP/170] 04:09:41, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0, Push 637
2:112.1.1.1.10:1::10200::90:e2:ba:d1:e2:30::10.0.2.112/304 MAC/IP
                *[BGP/170] 00:19:07, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0, Push 637
3:112.1.1.1.10:1::10100::112.1.1.10/248 IM
                *[BGP/170] 04:09:42, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0
3:112.1.1.1.10:1::10200::112.1.1.10/248 IM
                *[BGP/170] 04:09:42, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0
4:112.1.1.1.10:0::0190e2bad1e230000f00:112.1.1.10/296 ES
                *[BGP/170] 02:25:53, localpref 100, from 112.1.1.10
                  AS path: 65002 I, validation-state: unverified
                > to 172.16.12.1 via ae0.0
4:112.1.1.1.10:0::0190e2bad93064000f00:112.1.1.10/296 ES
                *[BGP/170] 02:25:54, localpref 100, from 112.1.1.10
```

```
                    AS path: 65002 I, validation-state: unverified
                  ] to 172.16.12.1 via ae0.0

{master:0}
root@qfx1]
```

In the CLI output above, we notice that there are two Type-1 EVPN routes per host, these are AD/ESI and AD/EVI routes. the AD/ESI route advertises the Ethernet segment to which the host is connected and the AD/EVI route advertises the remote EVPN to which the route belongs.

By looking at the ESIs in these routes and recalling what we explained above about how are these Type-1 ESI are constructed, we can identify that the 1st and 5th Type-1 routes are relative to Host-2, and that the 2nd and 6th Type-1 routes are relative to Host-1.

Now, how about the 3rd and 4th routes? Let's dissect the ESI of these routes and figure it out.

ESI "050000fdea0000277400" can be deconstructed as follows:

- 0x05 : Autonomous System Number (ASN) based ESI
- 0x0000fdea : ASN = 65002 (decimal)
- 0x00002774 : Local Discriminator = 10100 (decimal). In this case, the VNI is used as a Local Discriminator
- 0x00 : the last octet is always zero

This Type-1 route is in fact relative to the Virtual Gateway that was auto-derived from the EZ-LAG configuration. by default, EZ-LAG configures IRB interfaces with a virtual gateway, but there are CLI knobs available to change that into anycast mode. Those commands are:

```
set services evpn global-parameters anycast-mac [mac_address]
set services evpn evpn-vxlan irb [irb_instance] use-anycast-address
```

We verified by testing, that when this mode is used, the 3rd and 4th Type-1 routes above do not show up anymore.

1.6 Conclusion

In this post, we explained EZ-LAG feature in details. This new feature was introduced by Juniper as an alternative to MC-LAG and based on EVPN-VXLAN technology. We showed that the configuration of this feature has been made much simpler than ESI-LAG, which requires a full-fledged EVPN-VXLAN configuration. EZ-LAG hides the complexity of an ESI-LAG configuration while still providing the same feature and the same service. The knobs supported by EZ-LAG CLI allow the user to tailor the configuration and avoid conflicts with existing one for other features on the same switch.

1.7 Useful links

- Understanding Multi-Chassis Link Aggregation Groups: https://www.juniper.net/documentation/us/en/software/junos/mc-lag/topics/concept/mc-lag-feature-summary-best-practices.html
- Commit Scripts Overview: https://www.juniper.net/documentation/us/en/software/junos/automation-scripting/topics/concept/junos-script-automation-commit-script-overview.html

1.8 Glossary

- AD : Auto Discovery
- ASN : Autonomous System Number
- BGP : Border Gateway Protocol
- CLI : Command Line Interface
- ESI : Ethernet Segment Identifier
- EVI : Ethernet VPN Instance
- EVPN : Ethernet Virtual Private Network
- LAG : Link Aggregation Group, aka bond or port channel
- MC-LAG : Multi-Chassis LAG
- OSPF : Open Shortest Path First
- VNI : Virtual Network Identifier
- VXLAN : Extended Virtual Local Area Network

1.9 Acknowledgments

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